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From Individual Balance Control to Collective Dynamics in Ultra-Dense Crowds

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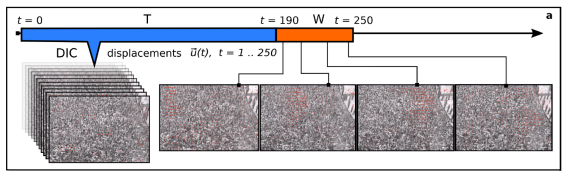
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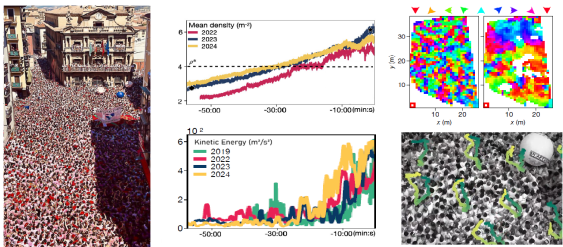


Collective behaviour in ultra-dense crowds: Empirical evidence

- ▶ **Propagating density waves** Oasis Concert, 60 000 attendees (2005)



- ▶ **Chiral rotations** San Fermín Festival & Love Parade Duisburg (2010)

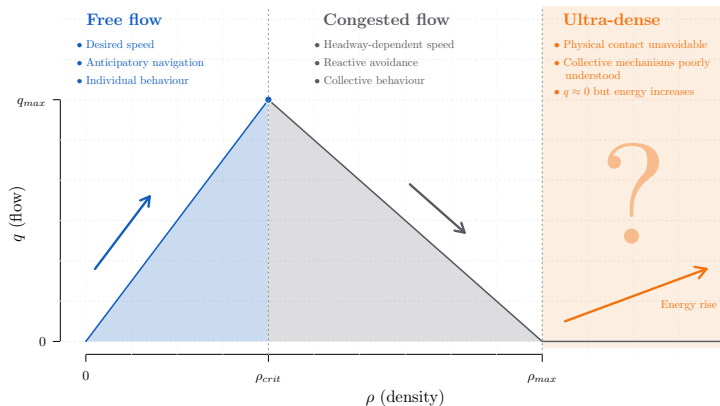


What in pedestrian behaviour can explain such large and potentially dangerous patterns?

Could they arise from pedestrian biomechanics and individual balance control mechanisms?



Pedestrian models



- ▶ Classical modelling concepts no longer hold in ultra-dense situations
- ▶ Dynamics mainly governed by physical contact forces and biomechanics



► Microscopic approaches

Pairwise contact forces

- 2000 Helbing & al. *Nature* **407**:487 – Body contact and friction; seminal dense crowd model
- 2013 Silverberg & al. *Phys Rev Lett* **110**:228701 – Vicsek-like model with algebraic contact and passive/active agents [↗](#)
- 2015 Kim & al. *Vis Comput* **31**:541 – ORCA collision avoidance model with contact forces
- 2016 Bottinelli & al. *Phys Rev Lett* **117**:228301 – Algebraic repulsion; reproduces high-density structural patterns

► Hybrid approaches

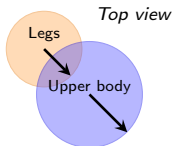
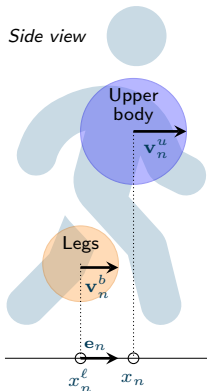
Kernel-based and mean-field continuum models

- 2009 Narain & al. *ACM Trans Graph* **28**:122 – Contact via incompressibility constraint
- 2021 Van Toll & al. *Comput Graph* **98**:306 – Smoothed particle hydrodynamics up to extreme densities
- 2025 Gu & al. *Nature* **638**:112 – Frictional and weathercock terms; collective oscillations in massive crowds

⇒ All models remain 2D and contact-based: **biomechanical balance control is neglected**



Two-level pedestrian model



$$\left\{ \begin{array}{l} \dot{\mathbf{x}}_n = \mathbf{v}_n \\ \dot{\mathbf{x}}_n^\ell = \mathbf{v}_n^\ell \\ \mathbf{e}_n = \frac{\mathbf{x}_n - \mathbf{x}_n^\ell}{\|\mathbf{x}_n - \mathbf{x}_n^\ell\|} \\ \dot{\mathbf{v}}_n = \underbrace{\lambda_u (v_u \mathbf{e}_n - \mathbf{v}_n)}_{\text{Unbalancing}} - \underbrace{\lambda \mathbf{v}_n}_{\text{Damping}} - \underbrace{\sum_{m \neq n} \nabla V(\mathbf{x}_n - \mathbf{x}_m)}_{\text{Upper body interaction}} \\ \dot{\mathbf{v}}_n^\ell = \underbrace{\lambda_b (v_b \mathbf{e}_n - \mathbf{v}_n^\ell)}_{\text{Balance recovery}} - \underbrace{\sum_{m \neq n} \nabla V_\ell(\mathbf{x}_n^\ell - \mathbf{x}_m^\ell)}_{\text{Legs interaction}} \end{array} \right.$$

Parameters:

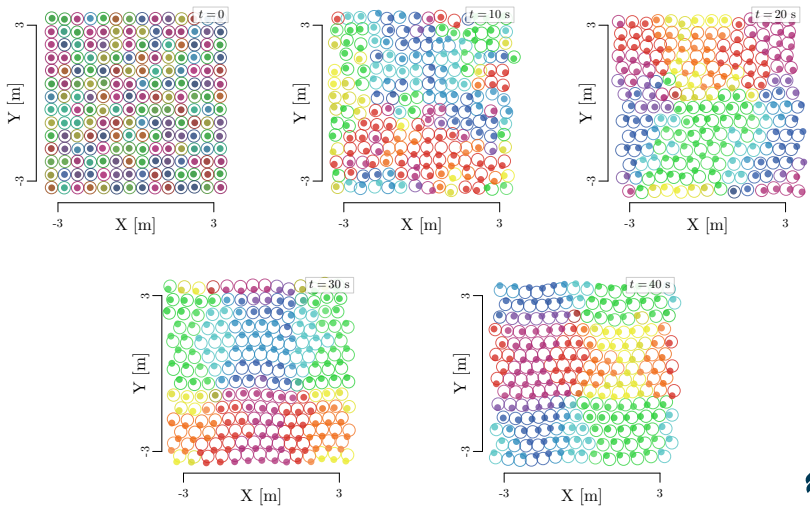
v_b, v_u	Balancing/unbalancing speed
λ_b, λ_u	Balancing/unbalancing rates
V, V_ℓ	Repulsive interaction potentials
λ	Damping rate



Simulation results

Large unbalancing speed: **Density wave pattern**

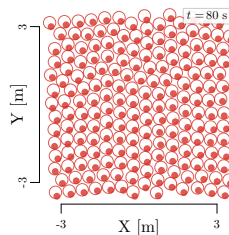
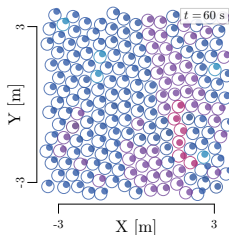
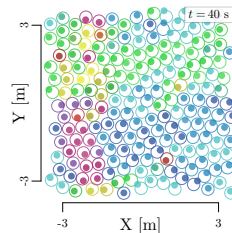
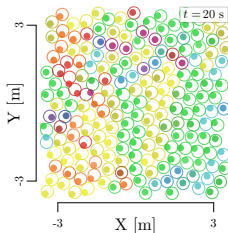
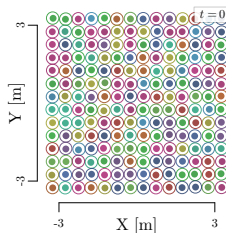
$\lambda_b = 1 \text{ s}^{-1}, \lambda_u = 0.5 \text{ s}^{-1}, v = 1 \text{ m/s}$



Simulation results

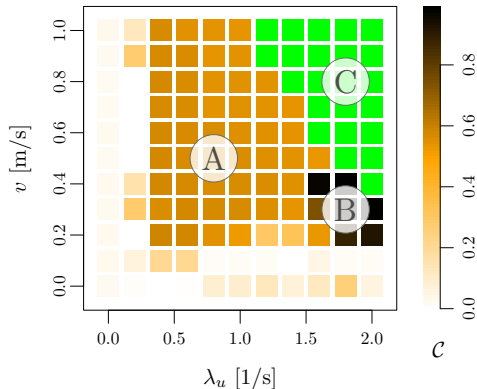
Large unbalancing rate: **Chiral rotation**

$$\lambda_b = 0.5 \text{ s}^{-1}, \lambda_u = 1 \text{ s}^{-1}, v = 0.2 \text{ m/s}$$



Phase diagram

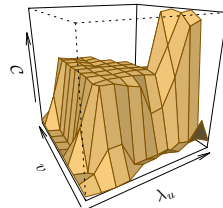
Velocity correlation $C = \frac{1}{N} \sum_n \frac{1}{N_n} \sum_{m \in D_n} \frac{\mathbf{v}_n \cdot \mathbf{v}_m}{|\mathbf{v}_n| |\mathbf{v}_m|}$, $\lambda_b = 1 \text{ s}^{-1}$



A: Density Wave

B: Chiral rotation

C: Disordered



Main results

- ▶ Minimal **two-level pedestrian model** (legs + upper body) for ultra-dense crowds that captures the basics of **balance–unbalance dynamics**
- ▶ Reproduces **coexisting collective modes** of density waves and chiral rotations
→ **Robust dynamical regimes** with clear phase boundaries
- ▶ Emergence from **local interactions** and **two-level biomechanical coupling**

Key insight

- ▶ Crowd $\approx$ network of **coupled inverted pendulums**
- ▶ **Biomechanical balance control** is the dominant organizing principle
→ Details of contact forces are **secondary**

Outlook

- ▶ Extend model: nonlinearities, heterogeneity, falling mechanisms
- ▶ **Calibration & validation** with real data



Thank you for your attention !

Preprint [arXiv:2601.05867](https://arxiv.org/abs/2601.05867) • T Chatagnon et al.
When legs and bodies synchronize: Two-level
collective dynamics in dense crowds

Videos and code available at
zenodo.org/records/19604515

Online simulation platform
[\[at\].github.io/Two-Level-Pedestrian-Model](https://[at].github.io/Two-Level-Pedestrian-Model)

Download slides



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